----STRINGS USING FUNCTIONS----

1. ***WAP that takes string from the user and pass it to function. The function finds and returns the number of character in the string.***

```
#include<stdio.h>
#include<conio.h>
int count(char [ ]);

int main(){

        char str[100];
        int length;
        printf("Enter any String:");
        gets(str);
        length=count(str);
        printf("\nThe number of character is:%d", length);
   getch();
   return 0;
}

int count(char s[ ]){
        int i=0;
        while(s[i]!= '\0'){
                i++;
        }

return i;
}
```

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2. WAP in C to read a string and display it in reverse order.

```
#include<conio.h>
#include<stdio.h>
void reverse(char[ ] );

int main(){
        char str[100];
        printf("Enter any String:");
        gets(str);
```

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```
        reverse(str);
   getch();
   return 0;
}

void reverse (char s[ ] ){
        int length=0, i;
        while(s[length]!= '\0'){
                length++;
        }

        for(i=length-1;i>=0;i--){
                printf("%c", s[i]);
        }
}
```

3. ***WAP that reads string form user and uses a user defined function to copy the contents of the read string into another character array changing lower case letters to upper case and upper case letters to lower case.***

```
#include<stdio.h>
#include<conio.h>

void change_and_copy(char[ ], char [ ]  );
int main(){
        char str1[100], str2[100];
        printf("Enter any String:");
        gets(str1);
        change_and_copy (str1, str2);
        printf("\nThe copied string:%s", str2);
    getch();
   return 0;
}

void change_and_copy (char s1[ ] , char s2[ ] ){
        int i;
        for(i=0; s1[i]!= '\0'; i++){
                //converting lower to uppercase
                if(s1[i]>='a' && s1[i]<='z'){
                        s2[i]=s1[i]-32;
                }
```

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```
            //converting upper to lowercase
            else if(s1[i]>='A' &&s1[i]<='Z'){
                s2[i]=s1[i]+32;
            }
            //for digits and symbols
            else{
                s2[i]=s1[i];
            }
        }
        //adding null character to end of string 2
        s2[i]='\0';
}
```

4. **WAP that reads string form user and uses a user defined function check whether a given character is present in a string or not and find its position**

```
#include<stdio.h>
#include<conio.h>
void search(char [ ] , char );
int main(){
        char str[100], ch;
        printf("Enter any String:");
        gets(str);
        printf("Enter character to search:");
        scanf("%c", &ch);
        search(str, ch);
   getch();
   return 0;
}
void search(char s[ ], char ch ) {
        int i, flag=0;
        for(i=0; s[i]!= '\0'; i++){
            if(s[i]==ch || s[i]==ch+32 || s[i]==ch-32){
                flag++;
                printf("\n%c is present at position:%d",ch, i);
            }
        }
        if(flag==0){
            printf("\n%c is not present in the string",ch);
        }

}
```

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5. WAP to check frequency of a given character in a string.

```
#include<stdio.h>
#include<conio.h>
void search(char [ ] , char );
int main(){
        char str[100], ch;
        printf("Enter any String:");
        gets(str);
        printf("Enter character to search:");
        scanf("%c", &ch);
        search(str, ch);
   getch();
   return 0;
}

void search(char s[ ], char ch ) {
        int i , flag=0;
        for(i=0; s[i]!= '\0'; i++){
                if(s[i]==ch || s[i]==ch+32 || s[i]==ch-32){
                    flag++;
                }
        }
        if(flag==0){
                printf("\n%c is not present in the string",ch);
        }
        else{
                printf("\nFrequency of %c in %s is:%d",ch, s, flag);
        }
}
```

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